MIL-P-50791C (AR)
8 MAY 1987
SUPERSEDING
MIL-P-50791B (AR)
30 October 1979

MILITARY SPECIFICATION

PROJECTILE, 105MM, TPDS-T, M737
METAL PARTS ASSEMBLY

This Specification is approved for use within the US Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements and quality assurance provisions for the fabrication of parts, assemblies and packing of projectile 105MM, TPDS-T, M737.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

PPP-T-60	- Tape, Packaging, Waterproof
PPP-T-76	- Tape, Packaging, Paper (for carton sealing)
PPP-F-320	- Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes
UU-C-282	- Chipboard
TT-C-490	- Cleaning Method and Pre-treatment of Ferrous Surfaces for Organic Coatings
PPP-B-636	- Boxes, Shipping, Fiberboard

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament Munitions and Chemical Command, Attn. AMSMC-QA, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY

MIL-B-117 - Bags, Sleeves and Tubing Interior Packaging
MIL-A-2550 - Ammunition, General Specification for
MIL-A-8625 - Anodic Coatings for Aluminum and Aluminum Alloys
MIL-A-13374 - Adhesive, Dextrin, For Use in Ammunition Containers
MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For
MIL-B-48570 - Band, Centering For Projectile 105MM Discarding Sabot
MIL-R-50792 - Rotating Band Tubing, Vulcanized Fiber
MIL-B-63454 - Band, Rotating, Nylon Glass-Filled for Projectile, 105MM, M737 TPDS-T
MIL-V-63118 - Vulcanized Rubber Obturator

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-1167 - Ammunition Data Cards

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS (See 6.10)

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

7548604 - Pallet (Boxed Type) For Projectiles and Projectiles Metal Parts
11738365- Projectile 105MM, TPDS-T, M737 Metal Parts Assembly
8790670 - Desiccant

(Copies of specifications, standards, handbooks, drawings, publications and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN NATIONAL STANDARD INSTITUTE

ANSI B46.1 - Surface Texture

(Application for copies should be addressed to American National Standards Institute (ANSI), 1430 Broadway, New York, NY 10018).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

E8 - Tension Testing of Metallic Materials
E18 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials
B117 - Salt Spray (FOG) Testing

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103).

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS Standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. Materials, parts and assemblies shall comply with the requirements specified on the drawings, specifications and the following:

3.2 Heat treatment or stress relief. Heat treatment or stress relief applied shall be uniform throughout each lot of components and shall be so conducted as to produce in each component the mechanical properties specified on the component drawing.

3.3 Mechanical properties. Mechanical properties shall comply with the requirements of the drawing or specification throughout the component.

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3.4 Rotating band, vulcanized fiber. Once rotating bands have been machined from tubing that has complied with the physical and mechanical property requirements of the component drawing and specification, and have been accepted for dimensional compliance no subsequent remachining to restore dimensions is permitted. No lubricants or coolants shall be used in any machining operation of the blanks or the machined bands.

3.5 Assembly precautions.

3.5.1 Cleanliness of components. To avoid malfunction of the metal parts assembly, it is essential that all components prior to assembly are perfectly clean and dry, particularly the rear taper of the subcaliber projectile assembly and the internal taper of the base.

3.5.2 Security of the subcaliber projectile. The subcaliber projectile assembly shall not rotate in the metal parts assembly when subjected to a torque of 50 ± 5 inch-pound (5.65 ± .56 newton-metre) in a counterclockwise direction (as viewed from the front of the metal parts assembly).

3.6 Surface preparation and application of protective finish. Where cleaning and surface pretreatment are required by component details, the process shall apply to all indicated surfaces regardless of contour configuration. The protective finish shall be as specified on the drawings and shall be uniformly applied to the surfaces designated by the drawings. Salt spray and other required testing shall be in accordance with drawings and specifications.

3.7 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.8 Proving ground functional tests. The metal parts assemblies shall comply with the following requirements when tested as specified in 4.5.6.

3.8.1 Metal parts security. There shall be no breakup of the subcaliber projectile in the gun bore or in flight, prior to impact. In the event of base or sabot (excluding petals) breakup, the lot shall be referred to the Contracting Officer for evaluation and for final determination of acceptability.

3.8.2 Flight. There shall be no evidence of erratic flight (see 6.2.3) greater than the numbers specified in Tables II and III during any of the proving ground tests conducted.

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3.8.3 Target-impact-dispersion. The pooled probable error (P.E.) shall be not greater than 0.25 mils horizontal and 0.22 mils vertical. See 4.5.6.

3.9 Workmanship. Requirements for workmanship for all components are as shown on the component drawing, in the specification and the following:

3.9.1 Material defects. Soundness of material shall be as required by the specifications and drawings. In addition, all components shall be free from cracks, splits, cold shuts, inclusions, porosity and any other similar defects.

3.9.2 Surface finish. Surface finish shall be in accordance with the drawing and the specifications referenced therein.

3.9.3 Protective coating. Protective coating shall be uniform and undamaged. See 3.6.

3.9.4 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required by the drawings and shall be in accordance with the provisions of MIL-A-2550.

3.9.5 Burrs. No component shall have a burr which might interfere with the assembly or functioning of the projectile or which might be injurious to personnel handling the item.

3.9.6 Foreign matter. All components shall be free of chips, dirt, grease, kerosene, corrosion or other foreign matter.

3.9.7 Discoloration. Discoloration is not allowed with the exception of centering bands and chromate stains or phosphated components.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of section 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system in quality program.

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The absence of any inspection requirements in this specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The following types of inspections shall be conducted on this item:

a. First Article Inspection
b. Quality Conformance Inspection

4.3 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the items in sample quantities as indicated.

Part Description	Drawing	Quantity
Centering band	See MIL-B-48570	
Obturator	See MIL-V-63118	
Rotating band tubing - fiber	See MIL-R-50792	
Rotating band - machined - fiber	11738369	10
Rotating band - nylon	See MIL-B-63454	
Ring, antifriction	11738381	10
Base	9323906	10
Sabot (prior to slotting)	11738368	5
Sabot (after slotting)	11738368	10
Body	11738371	10
Nose	11738372	10
Sabot assembly	11738367	10
Subcaliber projectile assembly	11738370	10
Metal parts assembly	11738365	100
Metal parts assembly (uncoated)	11738365	2

Raw material - See Table I - Section IV.

See Note 1 of Table I.

Ninety of the above finished metal parts assemblies shall be forwarded to a designated Government facility for loading and subsequent ballistic testing. The sample shall be manufactured using the same materials, equipment, processes, and procedures as will be used in regular production. All parts and materials, including packaging and packing shall be the same as used in regular production and shall be obtained from the same source of supply. A First Article sample or portion thereof, as directed by the procuring agency shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing processes, materials used, drawings, specifications or source of supply, such as to significantly affect product uniformity as determined by the Contracting Officer.

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4.3.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. See MIL-A-48078.

4.3.4 Process description and controls. The contractor shall prepare a written description of manufacture including process controls for the sabot (11738368), base (9323906), nose (11738372), and body (11738371) as specified in the contract. As a minimum, the process description shall include intermediate steps in press operations and heat treatment/stress relief used to manufacture these parts. Process controls and inspections, including destructive tests used to assure proper quality of those parts and the sample sizes and frequencies shall also be stated. The contractor's written plan of process and process controls shall be prepared in accordance with Data Item Description DI-P-1604.

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TABLE I. First article inspection.

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-50791C (AR)

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
	Projectile, 105MM, TPDS-T, Metal Parts Assembly	1	2	11738365
				NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
I	Components (See Note 1 & 2)				
	a. 11738381 - Ring, antifriction	10		3.1	4.4.2.1
	b. 11738369 - Rotating band	10		3.1	4.4.2.2
	c. 9323906 - Base	10		3.1	4.4.2.6
	d. 11738368 - Sabot (Prior to slotting)	5		3.1	4.4.2.7
	e. 11738368 - Sabot (After slotting)	10		3.1	4.4.2.8
	f. 11738371 - Body	10		3.1	4.4.2.4
	g. 11738372 - Nose	10		3.1	4.4.2.3
II	Sub-Assemblies (See Note 1 & 2)				
	a. 11738367 - Sabot assembly	10		3.1	4.4.2.9
	b. 11738370 - Subcaliber projectile assembly	10		3.1	4.4.2.5
III	Metal Parts Assembly - 11738365				
	a. Examination for defects and tests	12	(See Note 3)	3.1	4.4.2.10
	b. Ballistic test	90		3.8	4.4.3.3.1
Note 1	If the component or sub-assembly has a protective coating, half of the samples shall be submitted without protective coating & the other half with protective coating.				
Note 2	Components & sub-assemblies shall be submitted for examination for defects & tests in accordance to the para. ref. column. All inspections & tests must be performed 100%.				

NOTES:

3 Two samples shall be submitted uncoated.

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TABLE I. First article inspection.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
	Projectile, 105MM, TPDS-T, Metal Parts Assembly				11738365
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
IV	Raw Material (Used for Mechanical Properties Testing, Whenever the end Item can't be used.) The following shall be submitted with material available for the appropriate mechanical property testing: 5 ends of 5 different bars, after thermal treatment used in the manufacture of the sabot with a min. length of one foot. 5 ends of 5 different bars, after heat treatment used in the manufacture of the base, with a min. length of one foot 5 ends of 5 different bars after heat treatment used in the manufacture of the body with a minimum length of one foot 5 ends of 5 different bars after heat treatment used in the manufacture of the nose, with a min. length of one foot 5 blanks, used in the manufacture of the antifriction ring.				
V	Centering band (MIL-B-48570) Obturator (MIL-V-63118), Fiber Rotating band tubing (MIL-R-50792) or nylon rotating band (MIL-B-63454). - The first article provision stated in each of the above specifications for each item apply.				

NOTES:

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition one lot of metal parts assemblies shall consist of not more than 15,000 units with:

a. Centering bands (MIL-B-48570) from lots of one interfix from one manufacturer.

b. Obturators (MIL-V-63118) from lots of one interfix from one manufacturer.

c. Rotating band fiber tubing (MIL-R-50792) or nylon rotating bands (MIL-B-63454) from lots of one interfix from one manufacturer.

4.4.1.1 Heat treatment lots (if applicable). Heat treatment lots shall contain metal parts forgings from not more than one heat of steel that has been heat treated in the same heat treatment equipment, at the same temperature, in one unchanged continuous process. Lot size shall not exceed 2500 components. The heat treatment lot thus constituted shall be the basis for the selection of samples for the tension test hereinafter prescribed.

4.4.1.2 Stress relieved lots (if applicable). Stress relieved lots shall contain cold work metal parts from not more than one heat of steel that has been cold worked by the same process, utilizing the same tooling and equipment and stress relieved in the same furnace at the same temperature for the same length of time. Lot size shall not exceed 2500 components. The stress relieved lot thus constituted shall be the basis for the selection of samples for the tension test hereinafter prescribed.

4.4.1.3 Stock material lots (if applicable). Stock material lots shall contain machined or forged (if applicable) metal parts from not more than one heat of bar stock steel that has been machined by the same process, tooling, and equipment for the same length of time. Lot size shall not exceed 10,000 components. The stock material lot thus constituted shall be the basis for the selection of samples for the tension test hereinafter prescribed.

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4.4.1.4 Miscellaneous lots. Small lots (such as metal parts manufactured from abnormally small heats of steel or metal parts that have become separated from previously delivered lots, etc.) may at the discretion of the supplier be grouped into miscellaneous heat treatment, stress relieved or stock material lots as the case may be. Such lots shall be identified by miscellaneous lot code numbers and processed in accordance with 4.4.1.1, 4.4.1.2 or 4.4.1.3 as applicable. The miscellaneous lot thus constituted shall be the basis for the selection of samples for the tension test hereinafter prescribed.

4.4.1.5 Production lots. When it has been determined that bodies of heat treated, stress relieved, stock material or miscellaneous lots have complied with the mechanical properties requirements of the applicable drawing, they may be regrouped for assembly into production lots, to which will be assigned a production lot number. The production lot shall consist of projectile assemblies produced by one supplier under one contract in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision, under conditions of continuous production free from interruptions other than those due to the end of a shift, day, or work week. The production lot thus constituted shall be the basis for the selection of samples for functional and protective finish testing hereinafter prescribed.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plan. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

b. Tool control dimensions. Dimensions marked tool control shall be gaged during the First Article testing and once in every 8 hour shift of regular production and prior to any tooling change where destruction of components is necessary to inspect these dimensions. Measurements of the tool may be substituted provided the contractor has established a correlation between the tool dimensions and the component dimensions prior to the start of production. The contractor shall be responsible for maintaining adequate surveillance to assure that drawing requirements are being met throughout the life of the tool. Corrective action will immediately be instituted by the contractor when drawing requirements are not met and all material produced since the last acceptance inspection shall be rejected.

c. Surface texture. The roughness comparison specimens prescribed by ANSI-B 46.1 shall be used as a basis for surface roughness determination.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-50791C (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.1	Ring, Antifriction				11738381
					NEXT HIGHER ASSEMBLY
					11738365
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Thickness		0.40%	3.1	Gage
102	Variation in thickness		0.40%	3.1	Gage
103	Variation in width		0.40%	3.1	Gage
Minor					
201	Diameter, outside		0.65%	3.1	Gage
202	Diameter, inside		0.65%	3.1	Gage
203	Gap		0.65%	3.1	Gage
204	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-50791C (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.2	Rotating Band - Fiber (Machined)				11738369
					NEXT HIGHER ASSEMBLY 11738367
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Hardness		0.40%	3.3	4.5.1.1
102	True position of counterbore with pitch diameter of thread		0.40%	3.1	Gage
103	Runout of outside diameter with pitch diameter of thread		0.40%	3.1	Gage
104	Pitch diameter of thread		0.40%	3.1	Gage
105	Minor diameter of thread, max.		0.40%	3.1	Gage
106	Diameter outside		0.40%	3.1	Gage
107	Diameter of counterbore		0.40%	3.1	Gage
108	Profile outside		0.40%	3.1	Gage
109	Length to forward basic diameter		0.40%	3.1	Gage
110	Length to rear basic diameter		0.40%	3.1	Gage
Minor					
201	Depth of counterbore		0.65%	3.1	Gage
202	Length, total		0.65%	3.1	Gage
203	Thread damaged or not full (beyond permissable limits)		0.65%	3.9.4	Visual
204	Radii or chamfers missing or incorrect		0.65%	3.1	Visual
205	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-50791C (AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.2.3	Nose				11738372
					NEXT HIGHER ASSEMBLY
					11738370
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Tension test		Note 1	3.3	4.5.2.2
102	Macro-examination (Peripheral grain growth)		Note 2	3.3	4.5.3
103	Runout of forward ogive with diameter forward of thread		0.40%	3.1	Gage
104	Runout of mid ogive with diameter forward of thread		0.40%	3.1	Gage
105	Runout of rear end of ogive with diameter forward of thread		0.40%	3.1	Gage
106	Runout of rear diameter with diameter forward of thread		0.40%	3.1	Gage
107	Runout of rear face with diameter forward of thread		0.40%	3.1	Gage
Note 1.	After final machining, one nose from every heat treated lot or two tensile test specimens for every 2,000 pounds (907Kg) of heat treated aluminum alloy rod, shall be selected at random and subjected to this test. If any test specimen fails the test, the lot shall be rejected.				
Note 2.	One specimen from every heat treated lot or from every 2,000 pounds (907Kg) of heat treated aluminum alloy rod shall be subjected to this test. If any test specimen fails the test, the lot shall be rejected.				

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-50791C (AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.3	Nose				11738372
					NEXT HIGHER ASSEMBLY
					11738370
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
108	Runout of shoulder with diameter forward of thread		0.40%	3.1	Gage
109	True position of pitch diameter of thread with diameter forward of thread		0.40%	3.1	Gage
110	Pitch diameter of thread, min		0.40%	3.1	Gage
111	Major diameter of thread, min.		0.40%	3.1	Gage
112	Diameter, forward of thread		0.40%	3.1	Gage
113	Diameter, rear end		0.40%	3.1	Gage
114	Diameter, rear end of ogive		0.40%	3.1	Gage
115	Length from rear face to the beginning of full form thread		0.40%	3.1	Gage
116	Length from rear face to the end of full form thread, min.		0.40%	3.1	Gage
117	Length from rear face to shoulder		0.40%	3.1	Gage
118	Protective coating inadequate		4.5.4	3.6	4.5.4
119	Salt spray test		4.5.4.1	3.6	4.5.4.1
Minor					
201	Pitch diameter of thread, max.		0.65%	3.1	Gage
202	Length from rear face to end of full or imperfect thread, max.		0.65%	3.1	Gage
203	Length from shoulder to tip of ogive		0.65%	3.1	Gage
204	Radii or chamfer, missing or incorrect		0.65%	3.1	Visual
205	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-50791C (AR)

PARAGRAPH	TITLE		SHEET 1 OF 3		DRAWING NUMBER 11738371
4.4.2.4	Body				NEXT HIGHER ASSEMBLY 11738370
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Tension test		Note	3.3	4.5.2.1
102	Hardness		0.40%	3.3	4.5.1.2
103	Runout of inside diameter at the rear of large thread with outside diameter		0.40%	3.1	Gage
104	Runout of inside diameter forward of large thread with outside diameter		0.40%	3.1	Gage
105	Runout of forward outside diameter of body with outside diameter		0.40%	3.1	Gage
106	Runout of rear shoulder with outside diameter		0.40%	3.1	Gage
107	Runout of rear basic diameter with outside diameter.		0.40%	3.1	Gage
108	Runout of tracer cavity with outside diameter		0.40%	3.1	Gage
109	Runout of tracer boss with outside diameter		0.40%	3.1	Gage
110	True position of pitch diameter of large thread with inside diameter forward of large thread		0.40%	3.1	Gage
111	True position of tracer thread with tracer cavity		0.40%	3.1	Gage

NOTES: After final heat treatment, two bodies from each heat treated lot or six tensile test specimens from every 1,000 ft. of heat treated steel alloy bar, shall be selected at random and subjected to this test. If any test specimen fails the test, the lot shall be rejected.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-50791C (AR)

PARAGRAPH	TITLE		SHEET 2 OF 3		DRAWING NUMBER 11738371
4.4.2.4	Body				NEXT HIGHER ASSEMBLY 11738370
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
112	Pitch diameter of large thread, max.		0.40%	3.1	Gage
113	Minor diameter of large thread, max.		0.40%	3.1	Gage
114	Length from forward face to the beginning of full form thread - Large thread		0.40%	3.1	Gage
115	Length from forward face to the end of full form thread, min. - Large thread		0.40%	3.1	Gage
116	Pitch diameter of tracer thread, max.		0.40%	3.1	Gage
117	Minor diameter of tracer thread, max.		0.40%	3.1	Gage
118	Length from rear face to full form thread, min. - tracer thread		0.40%	3.1	Gage
119	Inside diameter, rear of large thread		0.40%	3.1	Gage
120	Inside diameter, forward of large thread		0.40%	3.1	Gage
121	Outside diameter, front face		0.40%	3.1	Gage
122	Outside diameter of body		0.40%	3.1	Gage
123	Diameter of tracer cavity		0.40%	3.1	Gage
124	Diameter of tracer boss		0.40%	3.1	Gage
125	Depth of large cavity		0.40%	3.1	Gage
126	Depth of tracer cavity		0.40%	3.1	Gage
127	Length from shoulder to bottom of large cavity		0.40%	3.1	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-50791C (AR)

PARAGRAPH	TITLE		SHEET 3 OF 3		DRAWING NUMBER 11738371
4.4.2.4	Body				NEXT HIGHER ASSEMBLY 11738370
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
128	Length from rear face to shoulder		0.40%	3.1	Gage
129	Length from front face to forward basic diameter		0.40%	3.1	Gage
130	Length from shoulder to rear basic diameter (if applicable)		0.40%	3.1	Gage
131	Angle of rear taper (if applicable)		0.40%	3.1	Gage
132	Threads damaged or not full		0.40%	3.9.4	Visual
133	Surface finish improper in rear taper (if applicable)		0.40%	3.9.2	Visual
Minor					
201	Pitch diameter of large thread, min.		0.65%	3.1	Gage
202	Pitch diameter of tracer thread, min.		0.65%	3.1	Gage
203	Length from rear face to end of full or imperfect thread, - tracer thread		0.65%	3.1	Gage
204	Length from front face to end of full or imperfect thread, max. - Large thread		0.65%	3.1	Gage
205	Depth of drill point, max (if applicable)		0.65%	3.1	Visual
206	Radii or chamfers, missing or incorrect		0.65%	3.1	Visual
207	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-50791C (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.5	Subcaliber Projectile Assembly	1 OF 1	11738370
			NEXT HIGHER ASSEMBLY: 11738365

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major					
101	Gap between nose and body		100%	3.1	Gage
102	Torque (before painting)		0.40%	3.1	4.5.5.1
103	Runout of rear taper with nose & tracer cavity		0.40%	3.1	Gage
104	Length from shoulder to rear basic diameter (if applicable)		0.40%	3.1	Gage
105	Angle of rear taper (if applicable)		0.40%	3.1	Gage
106	Length from rear basic diameter to forward basic diameter		0.40%	3.1	Gage
107	Paint or foreign matter in rear taper or rear face (shoulder)		0.40%	3.1	Visual
108	Protective coating inadequate		4.5.4	3.6	4.5.4
109	Surface finish in rear taper, improper (if applicable)		0.40%	3.9.2	Visual
110	Salt spray test	2		3.6	See Note
Minor					
201	Paint thickness		0.65%	3.6	Gage
202	Evidence of poor workmanship		0.65%	3.9	Visual

NOTE:

If any subcaliber assembly fails to meet all the requirements of the drawing and TT-C-490, the lot shall be rejected. The actual subcaliber assembly shall be used in this test.

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AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-50791C (AR)

PARAGRAPH	TITLE		SHEET 1 OF 3		DRAWING NUMBER 9323906
4.4.2.6	Base				NEXT HIGHER ASSEMBLY 11738365
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major					
101	Tension test		Note 1	3.3	4.5.2.3
102	Macro examination (Peripheral Grain Growth)		Note 2	3.3	4.5.3
103	Runout of inside taper with diameter forward of thread		0.40%	3.1	Gage
104	Runout of small inside diameter with diameter forward of thread		0.40%	3.1	Gage
105	Runout of counterbore face, rear of inside taper with diameter forward of thread		0.40%	3.1	Gage
106	Runout of front face of forward flange with diameter forward of thread		0.40%	3.1	Gage
107	Pitch diameter of thread, min.		0.40%	3.1	Gage
108	Major diameter of thread, min.		0.40%	3.1	Gage
109	True position of pitch diameter of thread with diameter forward of thread		0.40%	3.1	Gage
Note 1.	After final machining, one base from every heat-treated lot or two tensile test specimens for every 2,000 pounds (907Kg) of heat-treated aluminum alloy rod, shall be selected at random and subjected to this test. If any test specimen fails the test, the lot shall be rejected.				

NOTES:
2. One specimen from every heat-treated lot or from every 2,000 pounds (907Kg) of heat-treated aluminum rod, shall be subjected to this test. If any specimen fails the test, the lot shall be rejected.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-50791C (AR)

PARAGRAPH	TITLE		SHEET 2 OF 3		DRAWING NUMBER 9323906
4.4.2.6	Base				NEXT HIGHER ASSEMBLY 11738365
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
110	True position of diameter rear of thread with diameter forward of thread		0.40%	3.1	Gage
111	Diameter, forward of thread		0.40%	3.1	Gage
112	Diameter, rear of thread		0.40%	3.1	Gage
113	Diameter of counterbore, rear of inside taper		0.40%	3.1	Gage
114	Diameter of counterbore at front face		0.40%	3.1	Gage
115	Diameter, inside small		0.40%	3.1	Gage
116	Diameter of rear flange		0.40%	3.1	Gage
117	Diameter of forward flange		0.40%	3.1	Gage
118	Diameter of obturator seat		0.40%	3.1	Gage
119	Angle of inside cavity		0.40%	3.1	Gage
120	Profile of obturator seat		0.40%	3.1	Gage
121	Length from front face to full form thread		0.40%	3.1	Gage
122	Length from front face of forward flange to full form of thread, max.		0.40%	3.1	Gage
123	Length from rear face to basic diameter		0.40%	3.1	Gage
124	Length from front face of forward flange to basic diameter		0.40%	3.1	Gage
125	Length from rear counterbore to basic diameter		0.40%	3.1	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 3 OF 3		DRAWING NUMBER
4.4.2.6	Base				9323906
					NEXT HIGHER ASSEMBLY
					11738365
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
126	Length from basic diameter to front face		0.40%	3.1	Gage
127	Thickness of rear flange		0.40%	3.1	Gage
128	Thread damaged or not full		0.40%	3.9.4	Visual
129	Radius or chamfer missing or incorrect in the rear flange		0.40%	3.1	Visual
130	Protective coating inadequate		4.5.4	3.6	4.5.4
131	Salt spray test		4.5.4.1	3.6	4.5.4.1
Minor					
201	Length from front face of forward flange to imperfect thread, min.		0.65%	3.1	Gage
202	Depth of forward counterbore		0.65%	3.1	Gage
203	Radius at bottom of rear cavity, max.		0.65%	3.1	Gage
204	Location of slots		0.65%	3.1	Gage
205	Radii or chamfers missing or incorrect		0.65%	3.1	Visual
206	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-50791C (AR)

PARAGRAPH	TITLE		SHEET 1 OF 4		DRAWING NUMBER 11738368
4.4.2.7	Sabot (Prior to Slotting and Painting)				NEXT HIGHER ASSEMBLY 11738367
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Macro-examination (Peripheral Grain Growth)		Note 1	3.3	4.5.3
102	Tension test		4.4.3.1	3.3	4.5.2.4
103	Metal defective		0.40%	3.9.1	Visual
104	Runout of inside forward taper with outside diameter		0.40%	3.1	Gage
105	Runout of surface between thread and shoulder with outside diameter		0.40%	3.1	Gage
106	Runout of internal groove with external groove		0.40%	3.1	Gage
107	True position of forward inside diameter with inside forward taper		0.40%	3.1	Gage
108	Runout of diameter forward of internal thread with outside diameter		0.40%	3.1	Gage
109	True position of pitch diameter of internal thread with diameter forward of internal thread		0.40%	3.1	Gage
110	True position of internal thread counterbore with diameter forward of internal thread		0.40%	3.1	Gage

NOTES: 1. One specimen from every heat treated lot or from every 1300 pounds (590 Kg.) of thermally treated magnesium alloy extruded rod shall be subjected to this test. If any specimen fails this test, the lot shall be rejected.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 4		DRAWING NUMBER
4.4.2.7	Sabot (Prior to Slotting and Painting)				11738368
					NEXT HIGHER ASSEMBLY
					11738367
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
111	True position of pitch diameter of rear external thread with surface between thread and shoulder		0.40%	3.1	Gage
112	Perpendicularity of rear face with internal thread counterbore		0.40%	3.1	Gage
113	Pitch diameter of rear external thread		0.40%	3.1	Gage
114	Length from shoulder to full form thread, max. (rear external thread)		0.40%	3.1	Gage
115	Length from rear face to full form thread, max. (rear external thread)		0.40%	3.1	Gage
116	Pitch diameter of internal thread		0.40%	3.1	Gage
117	Length from face to forward full thread, min. (internal thread)		0.40%	3.1	Gage
118	Pitch diameter of centering band thread		0.40%	3.1	Gage
119	Major diameter of rear external thread, min.		0.40%	3.1	Gage
120	Minor diameter of internal thread, max.		0.40%	3.1	Gage
121	Major diameter of centering band thread, min.		0.40%	3.1	Gage
122	Diameter, outside		0.40%	3.1	Gage
123	Diameter of grooves in centering band area		0.40%	3.1	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-50791C (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.7	Sabot (Prior to Slotting and Painting)	3 OF 4	11738368
			NEXT HIGHER ASSEMBLY: 11738367

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
124	Diameter of outside groove at rear of centering band seat		0.40%	3.1	Gage
125	Diameter forward of internal thread		0.40%	3.1	Gage
126	Diameter forward of small inside taper		0.40%	3.1	Gage
127	Diameter of internal thread counterbore		0.40%	3.1	Gage
128	Diameter between rear external thread and shoulder		0.40%	3.1	Gage
129	Diameter of shoulder, forward of rotating band seat		0.40%	3.1	Gage
130	Diameter of centering band seat		0.40%	3.1	Gage
131	Thickness of wall at basic length		0.40%	3.1	Gage
132	Width between inside groove and outside groove		0.40%	3.1	Gage
133	Diameter of inside groove		0.40%	3.1	Gage
134	Angle, inside forward taper		0.40%	3.1	Gage
135	Thread damaged or not full		0.40%	3.9.4	Visual
136	Thread serrations missing or incorrect		0.40%	3.1	Visual
Minor					
201	Diameter of leading angle, front face		0.65%	3.1	Gage
202	diameter of large cavity		0.65%	3.1	Gage
203	Location of leading angle, front face		0.65%	3.1	Gage
204	Length from rear face to forward inside basic diameter		0.65%	3.1	Gage
205	Length from front face to centering band shoulder		0.65%	3.1	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-50791C (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.7	Sabot (Prior to Slotting and Painting)	4 OF 4	11738368
			NEXT HIGHER ASSEMBLY 11738367

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
206	Length from rear face to shoulder above rear external thread		0.65%	3.1	Gage
207	Location of outside body groove		0.65%	3.1	Gage
208	Length of shoulder, forward of rotating band seat		0.65%	3.1	Gage
209	Length from rear face to bottom of internal thread cavity, max.		0.65%	3.1	Gage
210	Length from shoulder to imperfect thread, min. (rear outside thread)		0.65%	3.1	Gage
211	Length, total		0.65%	3.1	Gage
212	Width of outside groove at rear of centering band seat		0.65%	3.1	Gage
213	Width of inside groove		0.65%	3.1	Gage
214	Depth of internal thread counterbore		0.65%	3.1	Gage
215	Radii or chamfers missing or incorrect		0.65%	3.1	Visual
216	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-50791C (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 11738368
4.4.2.8	Sabot (After Slotting and Painting)				NEXT HIGHER ASSEMBLY 11738367
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Salt spray test	2	Note	3.1	See Note
102	Slots in nose missing or incorrect		0.40%	3.1	Visual
103	Paint missing or inadequate in internal and external shear grooves		0.40%	3.1	Visual
Minor					
201	Paint thickness		0.65%	3.1	Gage
202	Primer or paint on rear face, or interior surfaces at rear of large taper		0.65%	3.1	Visual
203	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:
If any sabot fails to meet all the requirements of the applicable drawing and TT-C-490, the lot shall be rejected. The actual sabot shall be used in this test.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-50791C (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.9	Sabot Assembly				11738367
					NEXT HIGHER ASSEMBLY 11738365
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Length from rear face of sabot to rear of rotating band seat		0.40%	3.1	Gage
102	Gap between front of rotating band and sabot shoulder		100%	3.1	See Note
103	Insecure assembly		0.40%	3.1	Manual
104	Foreign matter		0.40%	3.9.6	Visual
105	Rotating band outside diameter, min.		0.40%	3.1	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:

A .002 inch feeler gage must not enter more than one fourth projectile circumference in total.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-50791C (AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.2.10	Metal Parts Assembly				11738365
					NEXT HIGHER ASSEMBLY
					9278499
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Gap between base & bottom of sub-caliber projectile		100%	3.1	Gage
102	Loose subcaliber projectile		4.4.3.2	3.5.2	4.5.5.3
103	Torque between base and sabot assembly		0.40%	3.1	4.5.5.2
104	Runout of outside diameter of sabot with tracer cavity and nose		0.40%	3.1	Gage
105	Runout of centering band with tracer cavity and nose		0.40%	3.1	Gage
106	Diameter of centering band		0.40%	3.1	Gage
107	Presence of gap between sabot & base		0.40%	3.1	Gage
108	Gap between centering band and sabot, max.		0.40%	3.1	Gage
109	Diameter of sabot forward of nomenclature, max.		0.40%	3.1	Gage
110	Chamfer of centering band, (if applicable)		0.40%	3.1	Gage
111	Sharp edges, nicks or burrs in the rear flange of the base (including the wrench holes) before assembling the obturator		0.40%	3.1	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-50791C (AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.10	Metal Parts Assembly				11738365
					NEXT HIGHER ASSEMBLY
					9278499
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
112	Obturator damaged or improperly assembled		0.40%	3.1	Visual
113	Ballistic Test		Table III		
114	Antifriction ring missing		0.40%	3.1	Visual
Minor					
201	Marking missing, incorrect or illegible		0.65%	3.1	Visual
202	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-50791C (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.11	Box, Before and After Sealing (See Note)				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Sleeve, desiccant bag or disc missing or damaged		0.65%	3.1	Visual
202	Improper assembly		0.65%	3.1	Visual
	After Sealing				
203	Insecure sealing		0.65%	3.1	Visual
204	Box corners not blunted		0.65%	3.1	Visual
205	Box damaged or gouged through all layers		0.65%	3.1	Visual
206	Marking missing, incorrect or illegible		0.65%	3.1	Visual

NOTE:

Applies only when projectiles have vulcanized fiber rotating bands.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-50791C (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.12	Box,After Inserting in a Barrier Bag (See Note)				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Barrier damaged		0.65%	3.1	Visual
202	Marking missing, incorrect or illegible		0.65%	3.1	Visual
203	Barrier bag improperly sealed		0.65%	3.1	Visual

NOTES:

Applies only when projectiles have vulcanized fiber rotating bands.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-50791C (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.13	Pallet Bulk Pack - Open (See Note)				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Separator discs missing		100%	3.1	Visual
Minor					
201	Separator discs not glued or improperly located		0.65%	3.1	Visual
202	Bottom carton, cover carton, divider, sleeve, tube or disc damaged or gouged through all carton layers		0.65%	3.1	Visual
203	Packing material missing or improperly assembled		0.65%	3.1	Visual/Manual
204	Projectiles loose		0.65%	3.1	Manual
205	Improper number of projectiles packed		0.65%	3.1	Visual

NOTES:

Only to be used when projectiles have nylon rotating bands.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-50791C (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 7548604
4.4.2.14	Pallet				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Strapping missing, loose, or broken		0.65%	3.1	Visual/Manual
202	Pallet damaged (unserviceable)		0.65%	3.1	Visual
203	Improperly packaged		0.65%	3.1	Visual
204	Marking missing, incorrect or illegible		0.65%	3.1	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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4.4.3 Testing. Testing is described in the First Article and Quality Conformance tables.

4.4.3.1 Tension test - sabot.

4.4.3.1.1 Sample size. After thermal treatment, one test specimen taken from two different bars representing 1,300 pounds (590 Kg.) maximum of magnesium alloy extruded rod shall be selected at random and subjected to this test to determine compliance with the mechanical properties requirements of the drawings and specifications.

4.4.3.1.1.1 Yield strength and elongation. Failure of any tension test specimen to comply with the yield strength or elongation requirements shall be cause for rejection of the lot.

4.4.3.1.1.2 Tensile strength. If both test specimens have a tensile strength equal to or greater than 38,000 psi (262 MPa), the lot shall be accepted. If any tension test specimen has a tensile strength below 37,500 psi (258.5 MPa), the lot shall be rejected.

4.4.3.1.1.2.1 Retest. Retest will be allowed when the tensile strength of the tension test specimen (one or both) falls between 38,000 psi (262 MPa) and 37,500 psi (258.5 MPa). One test specimen shall be taken from the failing bar or bars plus one test specimen from a different bar selected at random from the lot. If any retested specimen has a tensile strength below 38,000 psi (262 MPa), the lot shall be rejected. Also if any retested specimen fails the yield or elongation requirement, the lot shall be rejected.

4.4.3.2 Torque test - security of sub-caliber projectile. Torque tests shall be applied according to the following schedule:

a. 100% inspection - 100% inspection shall be conducted until 20,000 successive units have passed the test.

b. Tightened random inspection - Succeeding 20,000 units shall be randomly inspected at a rate of 25% of production.

c. Normal random inspection - All subsequent units shall be inspected randomly at a rate of 10% of production.

d. Failure of one or more units in either the tightened or normal random inspection phases will revert inspection back to the 100% phase described in a. above.

4.4.3.3 Ballistic test.

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4.4.3.3.1 First article sample. Quantities of projectile assemblies listed in Table II shall be ballistically tested in the phases indicated.

TABLE II. First article ballistic test summary.

	TEMP °F	TUBE WEAR	QTY	REQUIREMENT	DEFECT (6.2.4)	ACCEPT	REJECT
A	+70	New	15	a) 3.8.3-Target-Impact-Dispersion	Major	4.4.3.3.3	
				b) 3.8.1-Security	Special	0	1
				c) 3.8.2-Flight	Major	0	1
B	+70	Worn	15	a) 3.8.1-Security	Special	0	1
				b) 3.8.2-Flight	Major	0	1
C	+125	Mid-life	15	a) 3.8.1-Security	Special	0	1
				b) 3.8.2-Flight	Major	(Note)	
D	+125	Worn	15	a) 3.8.1-Security	Special	0	1
				b) 3.6.2-Flight	Major	(Note)	
E	-25	Mid-life	15	a) 3.8.1-Security	Special	0	1
				b) 3.8.2-Flight	Major	(Note)	
F	-25	Worn	15	a) 3.8.1-Security	Special	0	1
				b) 3.8.2-Flight	Major	(Note)	

Note: C & D combined, accept on 1 reject on 2.
E & F combined, accept on 1 reject on 2.

4.4.3.3.2 Production lot sample. The quantities of projectile assemblies listed in Table III shall be ballistically tested in the phases indicated.

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TABLE III. Ballistic test summary.

TEMP °F	TUBE WEAR	QTY		REQUIREMENT	DEFECT (6.2.4)	ACCEPT	REJECT
A +70	New	10	a)	3.8.3-Target-Impact Dispersion	Major	4.4.3.3.3	
			b)	3.8.1-Security	Special	0	1
			c)	3.8.2-Flight	Major	0	1
B +125	Worn	10	a)	3.8.1-Security	Special	0	1
			b)	3.8.2-Flight	Major	0 (Note)	2

Note: If one round fails to meet the flight requirement, condition ten additional rounds at 70°F and fire from a worn tube. Reject the lot if in this retest one round doesn't meet the flight requirement.

4.4.3.3.3 Probable error (P.E.) in the target-impact-dispersion test. Vertical and horizontal hit pattern results shall be expressed as probable errors in mils and shall be recorded on the firing record. If the pooled P.E. (see 6.1) of the test rounds fired multiplied by the appropriate factor (see 6.1.3) of Table IV exceeds the requirement, the lot shall be rejected.

TABLE IV. Factors for computing acceptance of P.E.

SAMPLE SIZE	DEGREES OF FREEDOM	FACTOR
9	7	.71
10	8	.72
15	12	.76
24	19	.80
25	20	.80

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See 6.9 and MIL-A-48078 Section 6.

4.5 Methods of inspection.

4.5.1 Hardness test.

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4.5.1.1 Rotating band - Fiber. After final machining and within an eight hour period before assembly to the mating component, one hardness test shall be performed, in accordance with the material specification, on the outside diameter of the rotating band. A threaded fixture shall be utilized to hold the rotating band.

4.5.1.2 Body. After final heat treatment and final machining, the hardness test shall be performed on the large outside diameter of the body. Two tests, 180 degrees apart, shall be made using the procedure prescribed by ASTM-E-18.

4.5.2 Tension test. At the option of the contractor the tension test may be performed on the heat bar or on the heat treated machine part. At no time can material, either in bars or machined parts, be re-heat treated once the tension test has been performed and accepted. The Government reserves the right to perform the following tension test, at any time, on the machined parts.

4.5.2.1 Body. Two tensile test specimens shall be taken longitudinally from the wall of each body, approximately 180 degrees apart. In addition, one transverse specimen shall be taken from the rear area of each body. If the heat treated bar is used, two longitudinal specimens (180 degrees apart) and one transverse specimen shall be taken from each of two different bars. The longitudinal specimens shall be taken one inch (25MM) from the centerline of the bar to the centerline of the specimens. All test specimens shall conform as closely as possible to the largest obtainable round type specimen described in ASTM E-8. Yield strength shall be determined by the offset method utilizing the procedures prescribed by ASTM E-8. Transverse yield strength shall be determined and recorded for information only.

4.5.2.2 Nose. Two tensile test specimens shall be taken longitudinally from the nose, approximately 180 degrees apart at mid-radius when measured from the large diameter. If the heat treated bar is used, one longitudinal specimen shall be taken from the center of each of two different bars. The specimens shall be taken 0.5 in. from the centerline of the bar to the centerline of the specimens. All test specimens shall conform as closely as possible to the largest obtainable round type specimen described in ASTM E-8. Yield strength shall be determined by the offset method utilizing the procedures prescribed by ASTM E-8.

4.5.2.3 Base. Two tensile test specimens shall be taken longitudinally from the wall of the base, approximately 180 degrees apart. If the heat treated bar is used, one longitudinal specimen shall be taken from each of two different bars. The specimen shall be taken one inch from the centerline of the bar to the centerline of the specimen. All test specimens shall conform as closely as possible to the largest obtainable round type specimen described in ASTM E-8. Yield strength shall be determined by the offset method utilizing the procedures prescribed by ASTM E-8.

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4.5.2.4 Sabot. Test specimens selected in accordance with 4.4.3.1.1 and 4.4.3.1.1.2.1 shall be taken longitudinally approximately 1.9 in. from the centerline of the bar to the center line of the specimen. The specimens shall conform to the largest obtainable round specimen as described by ASTM-E-8. Yield strength shall be determined by the offset method utilizing the procedures prescribed by ASTM-E-8.

4.5.3 Macro examination (Peripheral grain growth). From each sample selected, a .50" (12.70mm) thick specimen shall be cut from the bar stock. After cutting the specimen face shall be prepared for macroetch by grinding and then etched in a suitable reagent to reveal grain structure. Each specimen shall be examined to determine compliance with drawing requirements. See 6.5.

4.5.4 Protective coating. Procedures for testing, the cleaning of components, and the application of protective coatings shall be in accordance with the criteria for testing as specified on component drawings and in the specifications referenced therein. If a salt spray is required by the specification, the test shall be conducted on components which have been manufactured in accordance with drawing requirements. Criteria for salt spray tests shall be as prescribed by ASTM-B117.

4.5.4.1 Salt spray test anodic aluminum components. The salt spray test shall be accomplished in accordance with MIL-A-8625.

Five samples shall be selected during first article and submitted to the salt spray test for a duration of 240 hours.

Two samples for every 5000 units shall be selected during regular production (each lot) for the salt spray test. These samples shall be exposed for 120 hours, beginning with the first lot produced and continuing until three consecutive lots have complied with requirements. After three consecutive lots have been accepted, the samples of all subsequent lots shall be exposed for 72 hours. If any lot fails, the salt spray samples of the following lots shall be exposed for 120 hours until three consecutive lots have been accepted.

4.5.5 Torque tests.

4.5.5.1 Subcaliber projectile assembly. Adhesive shall be cured for 60 minutes minimum before torque testing. The assemblies shall then be mounted in a suitable holding device in such a manner that the body is prevented from turning. Torque shall be applied to the nose in the opposite direction of assembly. The assemblies must withstand the specified disassembly torque without any perceptible movement.

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4.5.5.2 Metal parts assembly. Assemblies shall be mounted in a suitable holding device in such a manner that the sabot is prevented from turning. Torque shall be applied to the base in the opposite direction of assembly. Assemblies must withstand the specified disassembly torque without any perceptible movement.

4.5.5.3 Security of subcaliber projectile. An appropriate fixture shall be assembled to the metal parts assembly in such a manner that no rearward pressure is applied to that assembly. The specified torque shall then be applied and observation shall be made to determine compliance with 3.5.2.

4.5.6 Proving ground test. (See 4.4.3.3.1 and 4.4.3.3.2)

4.5.6.1 Ballistic phases. Metal parts assemblies shall be subjected to ballistic testing at the phases indicated in Table II (First Article) and Table III (Production). The metal parts assemblies shall be assembled into complete rounds utilizing the service charge and fired from a tank mounted M68 gun.

4.5.6.1.1 Reference rounds. Ten reference rounds shall be fired in the target-impact dispersion phase (70° - New Tube) in the first article test and in the production lot test. Firing sequence in the first article test shall be: five reference rounds, fifteen test rounds, five reference rounds. Firing sequence in the production lot test shall be: five reference rounds, ten test rounds, five reference rounds.

4.5.6.1.2 Target-impact-dispersion firing sequence. The test and reference samples shall be fired in five round groups (or occasions) with no less than three minutes nor more than five minutes between rounds. If a lapse of more than thirty minutes occurs between group firings, or if there was an indication of a trend in the previous group's target impacts to move away from the center of the target, the gun will be re-zeroed on target center. Target impacts of the test rounds including reference and spotter rounds shall be measured to the nearest centimeter in both the horizontal and vertical distance from the target center. Each individual reading must be recorded on the firing record. No more than one lot acceptance test shall be performed on each target. Firing cadence of five round groups will not be disrupted because of velocity equipment failure. Target-impact-dispersion tests will not be performed when cross winds are in excess of 10 knots or when gusts exceed 5 knots.

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4.5.6.1.3 Record of firing data. Individual velocities and pressures shall be recorded for information only. Target-impact-dispersion patterns shall be recorded in every firing test. Spin rate data shall be collected for the following tests only:

a. First Article Test.

b. Production lot acceptance tests until five successive lots within each lot interfix have passed.

c. Retest, conducted after failure of lot acceptance test.

d. Production lot acceptance tests for reworked lots.

e. Special tests when specifically directed.

4.5.6.1.4 Flight. The observer will be located at an elevated point behind the firing vehicle with an unobstructed view of the line of flight to the target and shall document projectile flight. Any abnormality of flight will be described (corkscrewing, tumbling, yaw, etc.,) as well as target impact information (hit target, short by 100 meters, left of target, etc.). If a round misses the target with no abnormality of flight, a statement to that effect should also be made.

4.5.6.1.5 Target misses. Target miss coordinates will be estimated and recorded for all test phases along with pertinent flight observations. See 4.5.6.8.

4.5.6.2 Temperature conditioning. All rounds fired shall be temperature conditioned for 24 hours, minimum, at the specified temperatures before firing.

4.5.6.3 Photographic equipment. Side smear cameras shall be utilized by the Proving Ground to determine compliance with 3.8.1, 3.8.2 and 3.8.3. Side smear cameras shall be located at 22 $\pm$ 0.5 ft (6.71 $\pm$ .15m) and 105 $\pm$ 5 ft (32.00 $\pm$ 1.52m) away from the gun muzzle.

4.5.6.4 Gun tube gaging. Before each test, the tube used shall be stargaged, both land and groove diameters in both horizontal and vertical star-gage orientations at 25.10 in (637.54mm), 25.25 in (641.35mm), 25.50 in (647.70mm) and 26.00 in (660.40mm) from the rear face of the tube and at one inch (25.4mm) intervals throughout the remaining length of the tube. The star-gage measurements shall be part of the records. See 6.2 for tube wear definitions. The tube shall be cleaned before star-gaging.

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4.5.6.5 Target. Rounds shall be fired through a target, 20 ft x 20 ft minimum (6.10m x 6.10m minimum) placed at 2000 meters from the muzzle of the gun.

4.5.6.6 Aiming. After loading and locking a round in the chamber, the gun tube shall be traversed off target center, dropped down below target center, traversed to the opposite side of target center, elevated to target center elevation, and traversed onto target center. The distance traversed and dropped should be sufficient to remove the backlash from the system. If the sight is above, or beyond target center, do not reverse direction of gun tube movement, but repeat the sequence.

4.5.6.7 Spotting the target. Spotter rounds shall be fired at the beginning of each firing phase. These rounds shall be temperature conditioned at the same temperature of the representing test phase. Spotter rounds will be used to assure the weapon is zeroed on target center for all test phases indicated. A minimum of three consecutive target hits within one mil of target center is required, and the three hits shall be obtained using the same elevation and deflection weapon settings. After the target has been spotted for a shot group, every effort should be made to keep the elevation and deflection of the weapon the same for that shot group.

4.5.6.8 Retest. If for any reason the proving ground considers that test conditions have detrimentally affected the test results, additional cartridges as required shall be tested. In addition, a lot shall be retested (one or more phases) whenever any of the following conditions exist:

a. Both the reference rounds and the test samples fired in the target impact dispersion phase fail to meet the performance requirements (see 3.8.3) of this specification.

b. If one round, other than erratic flights, misses the target in a phase where target-impact-dispersion is a requirement, this phase shall be retested if the pooled PE of the remaining rounds multiplied by the appropriate factor (see 4.4.3.3.3) meets the requirements of 3.8.3. Two or more target misses by the test rounds shall reject the lot represented by the samples. Target miss coordinates will be estimated and recorded for all test phases, along with pertinent flight observations.

c. If one or more rounds, other than erratic flights, misses the target in any phase where target-impact-dispersion is not a requirement, that phase shall be retested. Results of retests for target misses shall be referred to CDR, AMCCOM, AMSMC-QAM.

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4.5.6.9 Temperature tolerances. The following temperature tolerances apply to this specification.

a. -25°F + 5°F (-32°C + 3°C)
b. +70°F ± 2.5°F (+21°C ± 1.5°C)
c. +125°F - 5°F (+52°C - 3°C)

4.5.6.10 Gun zeroing - boresight and superelevation. Actual superelevation of the gun shall be measured and recorded for all target dispersion test phases. This requires that the gun be zeroed at the beginning, during any breaks in excess of 30 minutes and at the end of target impact test phases, using a conventional collimated M26 Pye Watson boresight (or equivalent). Record gun elevation and azimuth which aligns the boresight zero on the target center. Then record elevation and azimuth data for all test and calibration phases. The gun superelevation is the elevation during the test phases minus the elevation which produced a muzzle boresight on target center. This data shall be obtained when using either a fixed mount or a service tank gun system. This information shall be recorded for information only and included in the ballistic test report.

5. PACKAGING

5.1 Preservation and packing.

5.1.1 Level A. N/A.

5.1.2 Level B.

5.1.2.1 Unit pack - Projectile with vulcanized fiber rotating band. All material shall be thoroughly dry at time of packaging. The maximum moisture content of the packing material shall be 9%. Each projectile shall be unit packed in a "B" flute, 275 lbs test corrugated fiberboard box. The dimensions of the box shall be 4 1/2" X 4 1/2" X 11 3/8" high. Each box shall be provided with an "A" flute, 175 lbs test corrugated fiberboard sleeve with inside dimensions 4 1/8" diameter or square X 7 3/4" high; a spiral or convolute wound chipboard tube 3 1/2" long with inside diameter 2.00 - .03" and wall thickness 0.15" minimum; and a chipboard disc 3.5" diameter X 0.25" thick. With box assembled and top end closed and sealed, insert sleeve into box. Position projectile, base end down, on suitable pedestal support; place tube over nose of projectile, place box over projectile, invert box holding projectile in place, insert a one unit general purpose desiccant bag (see Dwg 8790670) into a void space within box, place disc on base of projectile, close and seal box. Blunt all box corners using hand mallet. Insert box, base end down, into barrier bag 9 7/8" X 18 1/2" high. Heat-seal bag, hold down edges, and stamp "TOP" on top end of unit pack. Handle top end up only.

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5.1.2.1.1 Construction and material requirements.

5.1.2.1.1.1 Fiberboard box and sleeve. The construction and material shall comply with PPP-B-636, Type CF, Class-Domestic, Style RSC. The manufacturer's joint shall be secured by reinforced paper tape, metal staples, or stitching shall not average over 1 1/2" apart, center to center. The sleeve shall have a butted body joint made by fitting the edges of the panels to be joined closely together and securing them with reinforced paper tape not less than two inches wide. The tape shall be centered on the joint and shall extend its full length, or within 3/8" of full length, and shall adhere over not less than 90% of the entire area of contact with the fiberboard.

5.1.2.1.1.2 Barrier bag. The construction and material shall comply with MIL-B-117, Type II, Class E, Style 1.

5.1.2.1.1.3 Chipboard disc and tube. The material shall comply with UU-C-282. Build up to obtain specified thickness by securing plies with dextrin adhesive, commercial, using individual plies conforming to requirements of the chipboard specification.

5.1.2.1.2 Sealing requirements.

5.1.2.1.2.1 Fiberboard box. The flaps shall be secured with glue conforming to MIL-A-13374, applied over at least 75% of the contact areas between inner and outer flaps. When flaps are pulled apart after the glue has set, they shall show complete fiber failure in the glued area. Alternately, the flaps may be secured with a two inch wide tape conforming to PPP-T-76 or PPP-T-60, Type III or IV. The tape shall be centered over the seam formed by the closure of the outer flaps of the top and bottom and shall extend down over the end panels not less than two inches.

5.1.2.1.2.2 Barrier bag. After exhausting excess air within the bag, seal in accordance with manufacturer's instruction, with the seal located at the top edge of the bag. The seal shall comply with MIL-B-117. After the bag is sealed, there shall be no visible collapsing of the sidewalls of the box. The bag shall be in intimate contact with the box (i.e., the bag shall create a smooth and tight fit over the box). At the edges of the box, a slight amount of residual air is permissible.

5.1.2.2 Unit pack - Projectile with nylon rotating band. All material shall be thoroughly dry at time of packaging. The maximum moisture content of the packing material shall be 9%. Place a carton bottom (Fig 1) into a pallet (see 5.1.2.3.2). Place a separator (Fig 3) into the carton bottom. Position 14 dividers (Fig 4) on the separator in the carton bottom to form 64 egg-crate type enclosures centered over the discs on the separator. Insert a

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sleeve (Fig 6) into each enclosure until it contacts the separator. Place a projectile, base end down, into each enclosure, assuring that the obturator does not rest on the disc. Position a tube (Fig 5) over the nose of each projectile. Place a carton cover (Fig 2) over the carton bottom.

5.1.2.2.1 Construction and material requirements.

5.1.2.2.1.1 Carton bottom and cover. The carton bottom and cover shall be constructed in accordance with Figures 1 and 2. The material shall be corrugated fiberboard in accordance with PPP-F-320, Type CF, Class-Domestic, Variety DW, Grade 500, 0.25 ± .03 inch thick.

5.1.2.2.1.2 Separator. The separator, consisting of a base and 64 discs, shall be constructed in accordance with Figure 3. The material for the base shall be corrugated fiberboard in accordance with PPP-F-320, Type CF, Class-Domestic, Variety DW, Grade 500, 0.25 ± .03 inch thick. The material for the disc shall be chipboard in accordance with UU-C-282. The disc shall have a diameter of 3.50 ± .06 inches and shall be built up to a thickness of 0.25 ± .03 inch by securing individual plies of chipboard with dextrin adhesive, commercial. The discs shall be glued to the base with dextrin adhesive, commercial, spread over at least 50% of the contact area between disc and base. When the disc is pulled apart from the base after the glue has set, there shall be complete fiber failure in the glued area.

5.1.2.2.1.3 Divider. The divider shall be constructed in accordance with Figure 4. The material shall be corrugated fiberboard in accordance with PPP-F-320, Type CF, Class-Domestic, Variety DW, Grade 200, 0.25 ± .03 inch thick.

5.1.2.2.1.4 Tube. The tube shall be spiral or convolute wound, constructed in accordance with Figure 5. The material shall be chipboard in accordance with UU-C-282. The tube shall be built up to specified thickness by securing individual chipboard plies with dextrin adhesive, commercial.

5.1.2.2.1.5 Sleeve. The sleeve shall be constructed in accordance with Figure 6. The sleeve shall have a butted body joint made by fitting the edges of the panels to be jointed closely together and securing them with reinforced paper tape not less than two inches wide. The tape shall be centered on the joint and shall extend its full length, or within 3/8 inch of full length, and shall adhere over not less than 90% of the entire area of contact with the fiberboard. The sleeve material shall be corrugated fiberboard in accordance with PPP-F-320, Type CF, Class-Domestic, Variety SW, Grade 175, "A" flute.

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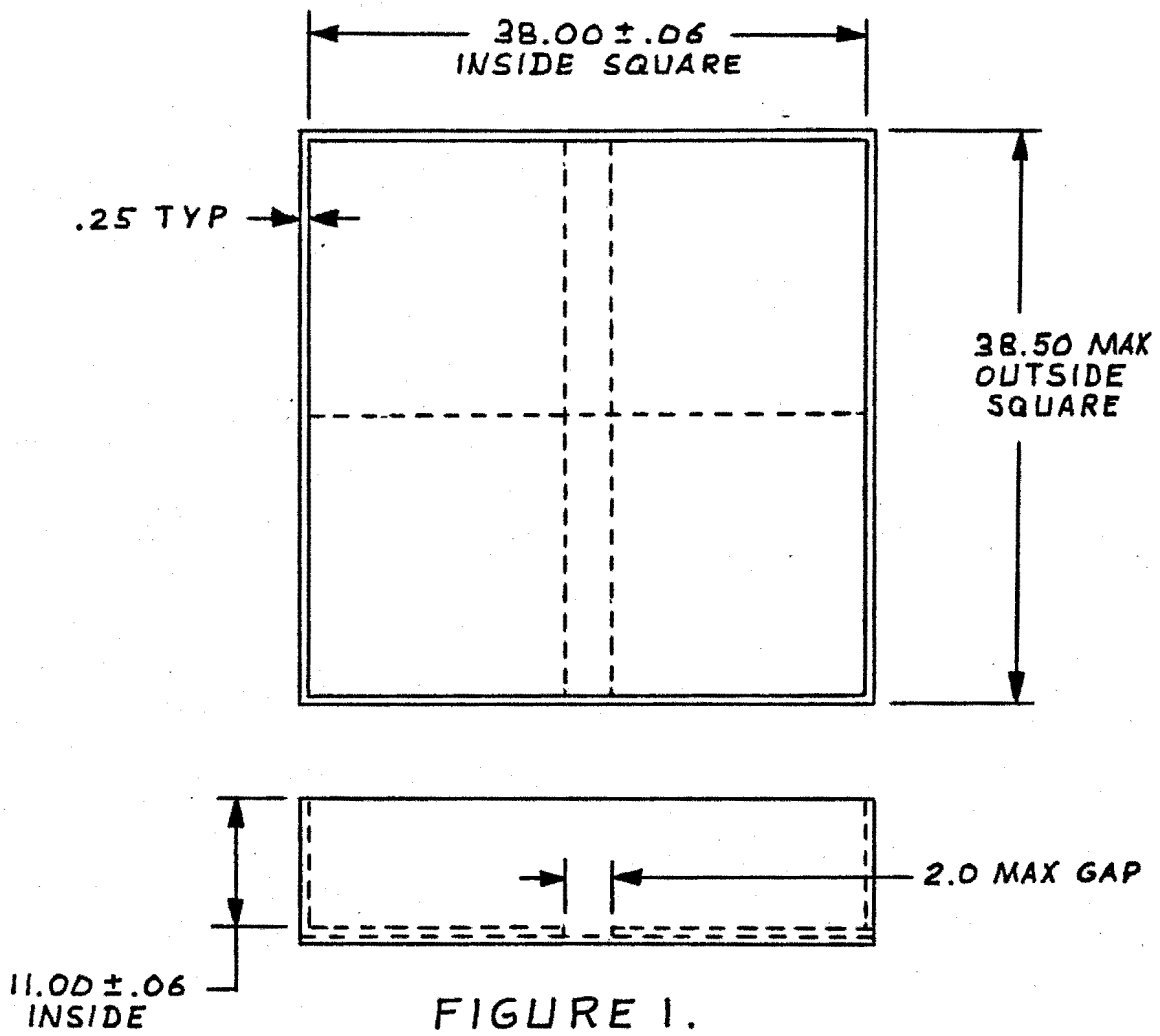


FIGURE 1.
CARTON BOTTOM

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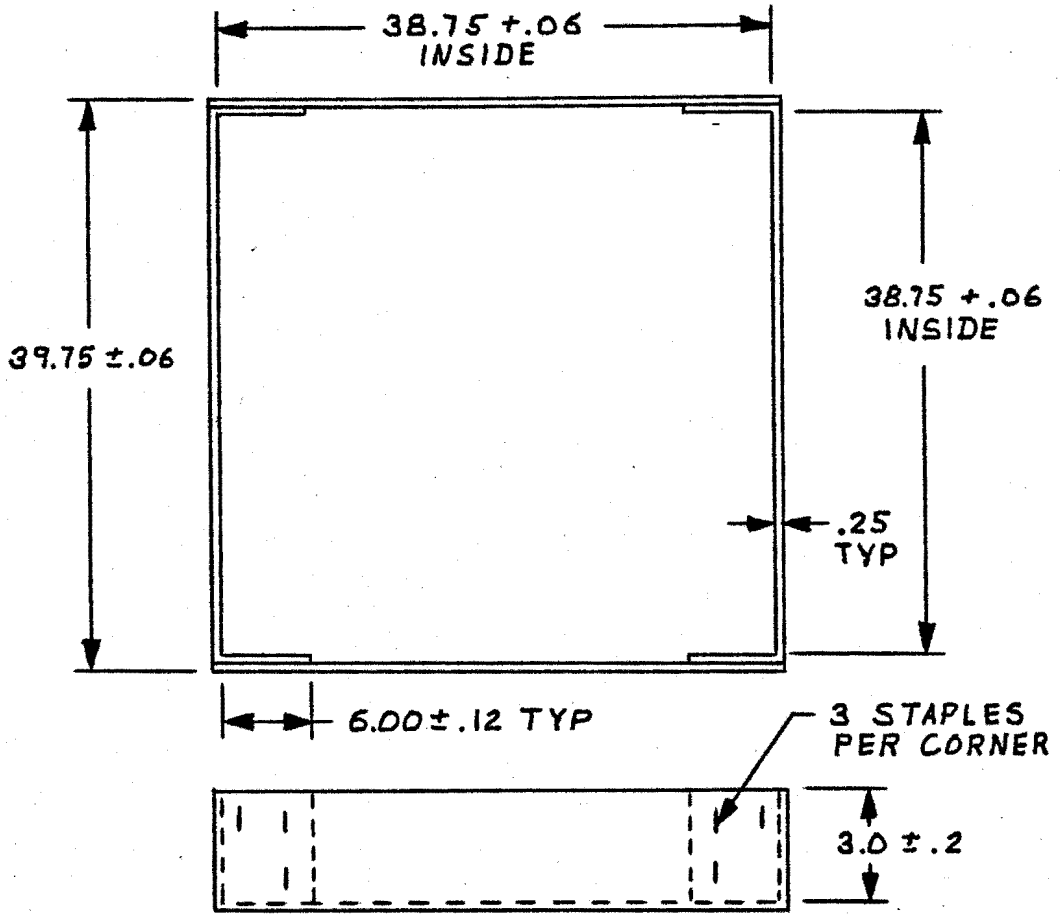


FIGURE 2.
CARTON COVER

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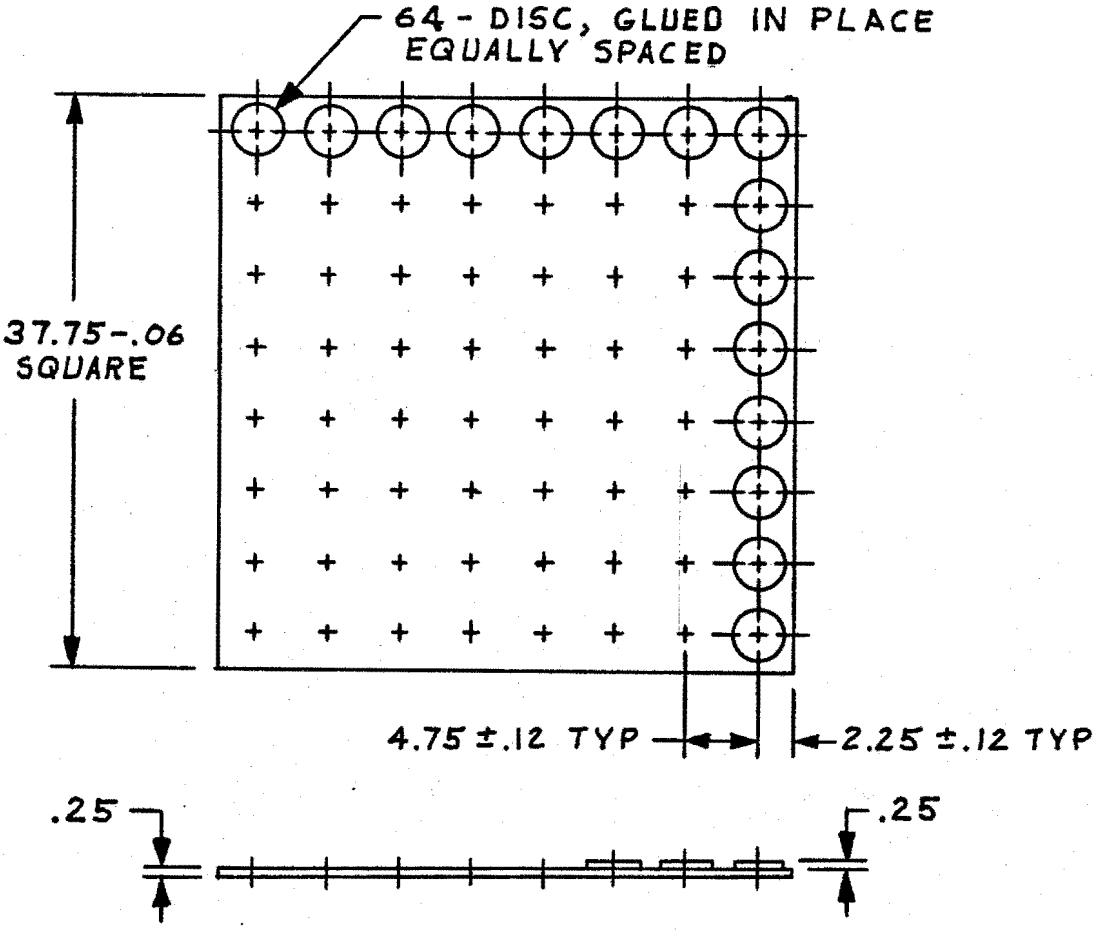


FIGURE 3.
SEPARATOR

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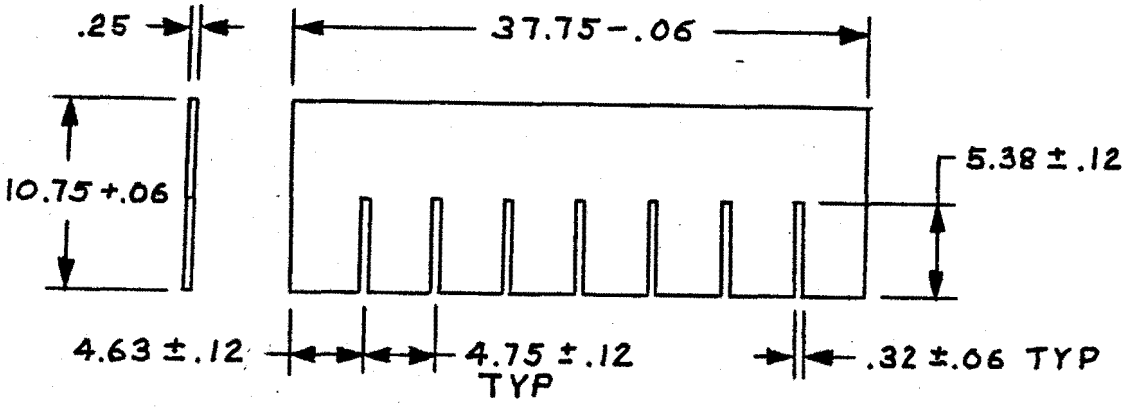


FIGURE 4. - DIVIDER

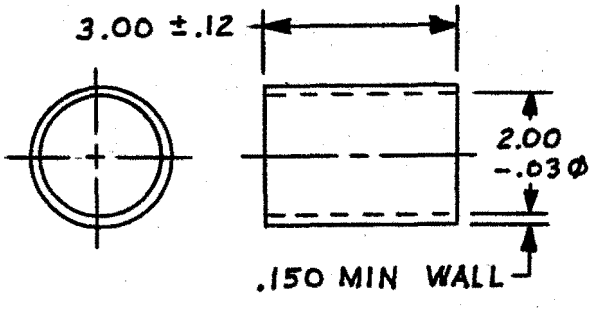


FIGURE 5. - TUBE

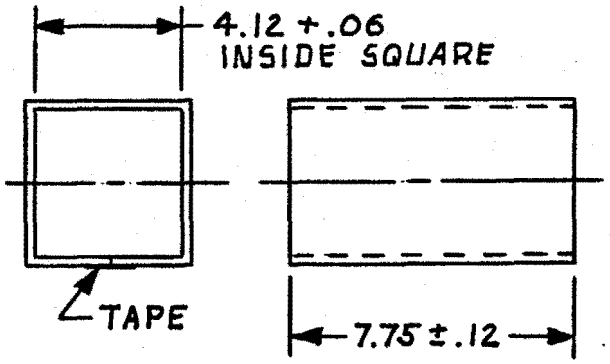


FIGURE 6. - SLEEVE

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5.1.2.3 Packing.

5.1.2.3.1 Projectile with vulcanized fiber rotating band. Projectiles shall be packed base ends down, on pallets manufactured in accordance with Drawing 7548604, using pallet dimensions as specified for Projectile, 105mm, APDS-T, M392. Additionally, 1/16 inch thick chipboard liner in accordance with UU-C-282 shall be placed along each pallet top and bottom to completely cover each surface. Lining of the inside corners is permitted to avoid damaging the unit packs during packing.

5.1.2.3.2 Projectile with nylon rotating band. Projectiles, in carton, shall be packed base ends down on pallets manufactured in accordance with Drawing 7548604, using pallet dimensions as specified for Projectile, 105mm, APDS-T, M392. Additionally, 1/16 inch thick chipboard liners in accordance with UU-C-282 shall be placed along the pallet sides and top, as required, to assure a tight fit and prevent movement of the carton within the pallet.

5.2 Marking. Marking shall be in accordance with MIL-STD-129. As a minimum, the following information is required:

National Stock No.
Quantity - Nomenclature
Lot No.
Pallet weight

5.3 Shipping. When projectiles from more than one lot are shipped as a carload, each lot must be kept separate and the division between lots clearly indicated to prevent mixing of lots in transit. The pallet information cards on two adjacent sides of the pallet will show lot numbers and quantities per lot.

6. NOTES

6.1 Method of calculating pooled PE. The individual PE results of the groups (5 samples in each group) of samples shall be pooled in accordance with the following formula:

$$PE_{pooled} = \sqrt{\frac{\sum \left[(N_i - 1) \quad PE_i^2 \right]}{\sum (N_i - 1)}}$$

Where: i = Number of groups
N_i = Number of samples in a group (target hits)
PE_i = Probable Error of the group

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6.1.1 Probable error calculations. The probable error of the group is calculated in accordance with the following equation:

$$PE_{mils} = \frac{(674.5)\ (SD)}{(X)\ (Rg)}$$

Where: SD = Standard deviation
Rg = Range in meters
X = Conversion factor
a) when target hits are measured in inches - 39.37
b) when target hits are measured in centimeters - 100.0

6.1.2 Standard deviation. The standard deviation of a sample is calculated in accordance with the following equation.

$$SD = \frac{N \quad X^2 - (\ X)^2}{N(N-1)}$$

Where: N = the number in the sample
X = the individual data point

6.1.3 Probable error factor. The factors shown in Table IV use the 95 percent upper confidence limit (A .05) and (N-1) degrees of freedom. To define (N_i-1) see 6.1.

6.2 Definitions

6.2.1 Primary wear. Wear measured in a vertical plane 25.25 inch (641.35mm) from the rear face of the tube.

6.2.2 Secondary wear. The maximum wear (both horizontal and vertical planes considered) down the tube from the 25.25 inch (641.35mm) location, as determined by the procedure of paragraph 4.5.6.4. (Secondary land and groove wear locations may not necessarily coincide.)

6.2.3 Erratic flight - Erratic flight is any abnormality in the subprojectile flight path, usually displayed by corkscrewing, tumbling or undulation.

6.2.4 Special defect - A special defect is a defect that judgement and experience indicate may, depending upon the degree of variance from the design requirement, result in unsafe conditions for persons using, maintaining or depending upon the product, or may prevent performance of the tactical function of a major end item.

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6.2.5 Gun tube wear limits. For the purpose of this test a new tube is defined as having no greater than .015 inches primary land wear and a midlife tube as having no greater than .030 inches primary land wear. A worn tube is defined as follows:

Primary Land Wear Limits	Secondary Land Wear Limits	Secondary Groove Wear Limits
Greater than 0.030 in. and less than .056 in.	Between 70% and 140% of primary land wear.	Between 80% and 190% of primary land wear.

6.3. Firing record information. Target impact plots, meteorological data and stargage measurements shall be included in all firing records.

Firing records shall include the serial numbers of all gun and fire control components, velocity reference/calibration ammunition component data where used, and gun to target azimuth.

Each round fired during the test shall be identified in the sequence of firing. For each round fired, the record shall contain deflection, elevation, time of firing, muzzle velocity, horizontal and vertical coordinates of impact, and flight characteristics.

In the firing tests where velocity, target impact dispersion, pressure and spin rate are recorded for information only, the results will be carefully analyzed to determine the quality of the ammunition produced. Evidence of excessive variation will result in an immediate investigation by the contracting officer.

6.4 Visual examination qualification. When compliance with requirements is in doubt as a result of visual examination, this characteristic may be measured or gaged to determine acceptability. The roughness comparison specimens prescribed by ANSI-B46.1 shall be used as a basis for surface roughness determination.

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6.5 Peripheral grain growth. Peripheral grain growth occurs as a result of achieving a critical degree of work hardening on the surface of the extrusion. Once this critical strain is achieved, recrystallization and grain growth occur. There is a very small gradient between unrecrystallized grains and recrystallized grains and grain growth is such that individual grains can be distinguished at one X. A properly prepared macro specimen suitably etched which had peripheral grain growth would be readily evident as a rim of extremely coarse grains. As a guide, grains in a zone of peripheral grain growth are approximately ten times larger than normal grains found elsewhere in the bar.

6.6 Salt spray test specimens. Salt spray samples shall be components randomly selected from each inspection lot. The contractor may use rejected components for salt spray samples. If the contractor uses rejected components, a request shall be forwarded to the contracting officer and approval received prior to use. A detailed plan describing the selection and processing of the sample shall accompany the request and shall serve as the basis for approval. The plan must clearly establish the relationship of the test specimens to the overall manufacturing process and lotting criteria which will apply to the components being accepted. Specifically, the plan shall address how test specimens are selected and identified; how they are processed at the same time and in the same way as the lot they represent; and how they are randomly introduced and distributed when the protective finish is applied. In addition, procedures shall be included for the contractor to require his quality control staff together with the resident DCAS quality assurance representative, to maintain surveillance over the process to ensure test specimens are not lost or given preferential treatment.

6.7 Ordering data. See MIL-A-48078.

6.7.1 Data Cards. Data cards shall be prepared for each lot in accordance with MIL-STD-1167.

6.7.2 Delivery of data to the technical activity. The following address is to be added to those listed in the contract (Form DD 1423) for the delivery of one copy, each, to the technical agency, of the Ammunition Data Cards prepared by the contractor for the duration of the contract: Commander, US Army Armament Research, Development and Engineering Center, ATTN: AMSMC-QAT-A(D), Picatinny Arsenal, New Jersey 07806-5000.

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6.9 Submission of test data. The following addresses are to be added to the Ballistic Test Request (BTR) for distribution of firing test reports to the technical activity: Commander, US Army Armament Research, Development and Engineering Center, ATTN: AMSMC-QAT-A(D), SMCAR-CCH-P and SMCAR-CCH-V, Picatinny Arsenal, New Jersey 07806-5000.

6.10 Submission of inspection equipment designs for approval. See MIL-A-48078. Submit equipment designs, as required to:

Commander
US Army Armament Research, Development and Engineering Center
ATTN: AMSMC-QAT-I(D)
Picatinny Arsenal, NJ 07806-5000

6.11 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as ARDC, ARRADCOM, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

6.12 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.13 Subject term (key word) listing.

Ammunition
Tank
Cannon
Cartridge
Projectile
Kinetic Energy

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1315-A757)

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